

Bi-branch network with region-wise dynamic convolution for image inpainting

Guangyuan Huang^a, Rong Huang^{a,b,*}, Shubo Zhou^{a,b} and Zhijie Wang^{a,b}

^aCollege of Information Science and Technology, Donghua University, Shanghai 201620, China

^bEngineering Research Center of Digitized Textile & Apparel Technology, Ministry of Education, Donghua University, Shanghai 201620, China

ARTICLE INFO

Keywords:

image inpainting
dynamic convolutions
region-wise convolutions
feature fusion

ABSTRACT

Recently, most image inpainting methods are based on convolutional neural networks, but the convolution kernels within networks are generally static, meaning that once networks are trained, their parameters remain fixed and do not adjust following inputs. As a result, static convolution kernels are intractable to accommodate the complexity and variability of the scenes. Additionally, they employ the same convolution kernels to simultaneously process features of missing and complete regions, ignoring the differences in feature distributions. In this paper, we propose a Bi-branch Network with Region-wise Dynamic Convolution (BNRDC), which consists of the upper and lower branches. The former is cored with Dynamic Convolution Kernel Prediction (DCKP) module and the latter is an inpainting backbone with an encoder-decoder architecture. DCKP only needs to predict a set of attention weights to combine multiple convolution kernels, rather than predicting all parameters of convolution kernels, to instantiate the Omni-Dimensional Dynamic Convolution (ODConv) kernels. At the bottleneck layer of the lower branch, we propose a Region Dynamic Convolution Attention (RDCA) module, which performs region-wise convolutions with the help of ODConv kernels to integrate long-range correlations. Furthermore, we design a Multi-Scale Feature Fusion (MSFF) module between the two branches, which fuse multi-scale features and deliver them to DCKP for predicting the ODConv kernels. Extensive experiments on Places2, CelebA and Dunhuang Challenge datasets demonstrate that the proposed method outperforms the state-of-the-art baselines with an average improvement of 3.2% for PSNR and 1.4% for SSIM, respectively. Ablation studies validate the effectiveness of each module, and visualized results indicate that RDCA has the ability to reduce the deviation between the features of the missing regions and ground-truth images.

1. Introduction

Image inpainting aims to fill in missing regions with the information of complete regions. It is a typical visual task with numerous applications including object removal [?], photo restoration [?] and image editing [?]. For image inpainting, the most crucial challenge is how to understand image semantics from complete regions and generate visually authenticity and semantically consistent content to fill the missing regions.

Existing image inpainting methods can be divided into two groups: traditional and deep generative methods. Traditional methods employ either based on partial differential equations [?] or patch-based approaches [?]. Due to the lack of understanding of high-level semantics and the inability to capture the overall structure of images, these methods perform poorly on images with complex scenes or large missing regions. The deep generative methods [?] employ Convolutional Neural Networks (CNNs) [?] to learn semantic information from data distribution and fill in missing regions with reasonable and realistic contents. Context Encoder proposed by Pathak et al. [?] is the first deep generative method based on convolutional networks for image inpainting, which takes an encoder-decoder architecture trained with an adversarial strategy. Methods [?] based on vanilla convolutions are not suitable for image inpainting because they treat missing

and complete pixels equally, leading to visual artifacts such as color discrepancies, blurriness, and noticeable edge responses surrounding the missing regions. In recent years, some methods [?] work around improving vanilla convolution for image inpainting, such as partial convolution [?], gated convolution [?] and region-wise convolution [?].

However, the convolution kernels adopted by the aforementioned methods are all static, meaning that their parameters remain fixed once the network is trained, regardless of inputs. This results in low generalization ability, especially when dealing with corrupted images with complex scenes. To address this issue, JPGNet [?] and MISF [?] introduce dynamic convolutions to accommodate the complex and diverse scenes. However, they are deficient in the following three aspects. Firstly, they treat the generation of dynamic convolution kernels as dense predictions of all parameters. In other words, they need to predict each parameter of the convolution kernel, resulting in low efficiency. Secondly, they adopt the same dynamic convolution kernel to simultaneously process features of missing and complete regions, overlooking the differences in feature distributions. Thirdly, the connection between their two branches is weak, and the features employed for predicting dynamic convolution kernels are limited.

In this paper, we propose a Bi-branch Network with Region-wise Dynamic Convolution (BNRDC), as shown in Figure 1, which is composed of the upper and lower

*Corresponding author

 rong.huang@dhu.edu.cn (R. Huang)

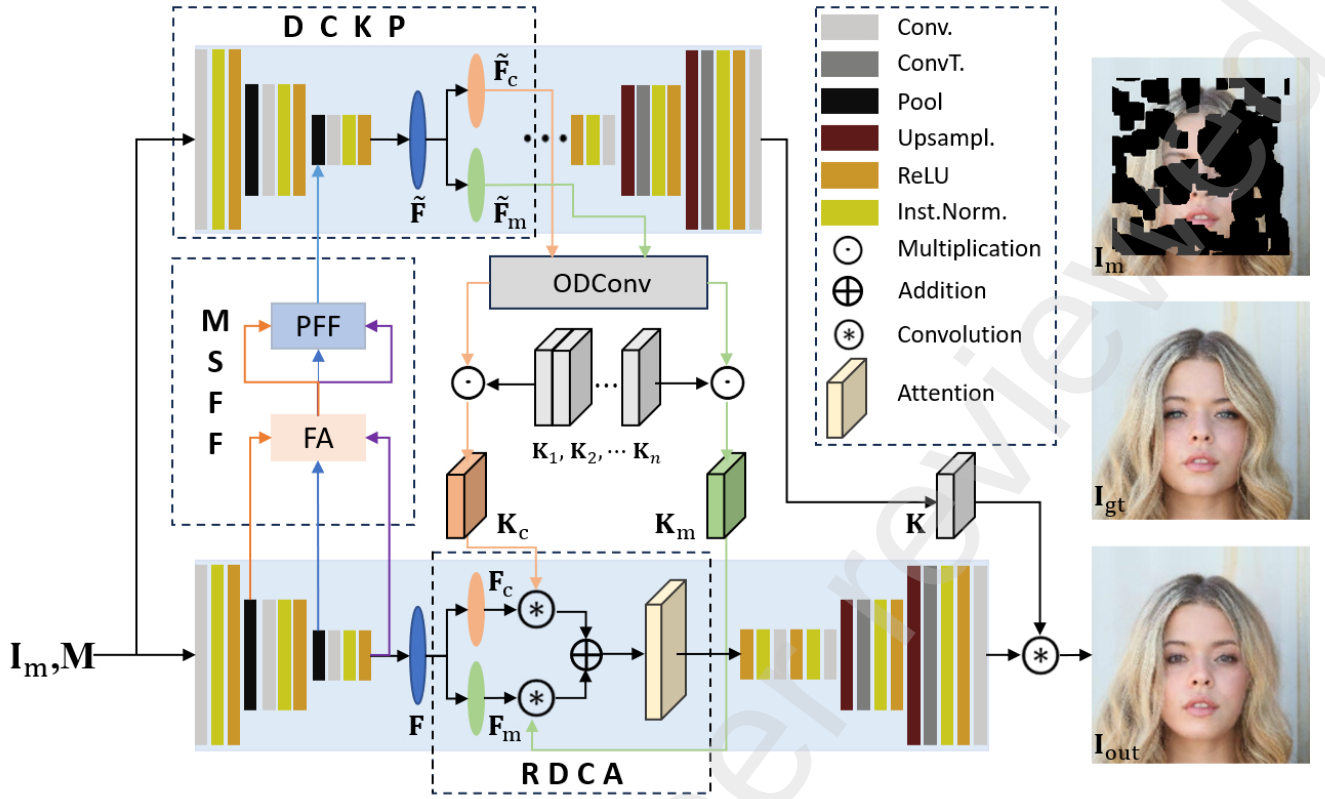


Figure 1: Overview of our BNRDC, which consists of the upper and lower branches and three modules: Multi-Scale Feature Fusion (MSFF), Dynamic Convolution Kernels Prediction (DCKP) and Region-wise Dynamic Convolution Attention (RDCA). DCKP and RDCA are the core components of the upper and lower branches, respectively. MSFF fuses multi-scale features from the lower branch encoder and provides them to DCKP. Based on the features fused by MSFF, DCKP predicts two dynamic convolution kernels for RDCA. The RDCA configures two dynamic convolution kernels to process features of missing and complete regions and fills in missing regions with the features of complete regions.

branches. The former is cored with the Dynamic Convolution Kernel Prediction (DCKP) module, while the latter serves as an inpainting backbone with an encoder-decoder architecture. Compared to predicting all parameters of dynamic convolution kernels, DCKP only needs to predict a set of attention weights to instantiate Omni-dimensional Dynamic Convolution (ODConv) kernels [?]. At the bottleneck layer of the lower branch encoder-decoder, we devise a Region Dynamic Convolution Attention (RDCA) module, which separately convolves features of missing and complete regions with two ODConv kernels to accommodate the differences in feature distributions. To provide sufficient feature information for predicting ODConv kernels, we design a Multi-Scale Feature Fusion (MSFF) module between the two branches, which fuses the multi-scale features and provides them for DCKP.

To demonstrate the superiority of our method, we conduct extensive experiments on the Places2 [?], CelebA [?] and Dunhuang Challenge [?] datasets. Qualitative and quantitative results demonstrate that our proposed method significantly outperform other state-of-the-art approaches [? ? ? ? ?] in image inpainting. Furthermore, we conduct visualization experiments to demonstrate that RDCA has the

capability to align features of missing regions closer to those of ground-truth images.

2. Related Works

The traditional methods in the field of image inpainting can be broadly categorized into two groups: based on partial differential equations [? ? ?] and patch-based [? ? ?] approaches. The former propagate contextual pixels from boundaries to the missing regions based on the isophote direction field. The latter iteratively search for the best fitting patches from complete regions to synthesize the contents of the missing regions based on nearest neighbor fields. However, these methods often fail to generate semantically meaningful and coherent content when dealing with corrupted images with large missing regions or complex structures. With the advancements in CNNs [?] and Generative Adversarial Networks (GANs) [?], deep generative methods [? ? ? ? ?] currently dominate image inpainting. These methods effectively extract meaningful semantics from corrupted images and recover visually plausible contents with high fidelity. One of the pioneering works in this field is the Context Encoder [?], which utilizes a deep neural network architecture consisting of an encoder-decoder pipeline to

predict missing regions. Building upon this, Iizuka et al. [?] introduce an additional discriminator to enforce global consistency. Shift-Net [?] combines the traditional patch-based idea into learning-based architecture by shifting the encoder features of the complete regions to the missing regions.

These deep generative methods are based on CNNs, vanilla convolutions indiscriminately process both complete and missing pixels, leading to color discrepancies and blurry inpainting results [?]. Some methods [?] focus on improving convolutions to repair missing regions with complete pixels and exclude the interference of missing pixels. Iizuka et al. [?] introduce dilated convolution [?] which captures rich contextual information by enlarging the receptive field of the convolution kernel, while keeping the number of parameters unchanged. To eliminate interference from missing pixels, Liu et al. propose PConv[?], which excludes missing pixels from convolution through an automatically updated binary mask. Based on PConv, Yu et al. propose gated convolution [?], which dynamically selects complete pixels with a learnable soft gated mask. Xie et al. [?] present a learnable attention map module to improve the mask update strategy of PConv, which learns feature re-normalization and mask-updating in an end-to-end manner. Yi et al. [?] develop a light weight gated convolutional layer, which employs a depth-wise convolution to compute the soft gated masks to reduce the number of parameters. Ma et al. [?] propose the region-wise convolutions which process features of complete and missing regions with two different convolution kernels.

Due to the inherent properties of CNNs, i.e., local inductive bias and spatial-invariant kernels, it is difficult for CNNs to model long-term correlations between distant contextual information and the missing regions. To address this issue, Yu et al. [?] propose the contextual attention which uses the features of complete patches as convolutional filters to process the missing patches. Wang et al. [?] design a multi-scale attention module, which fuses attention information of different patch sizes. PENNet [?] employs attention transfer networks which calculates attention scores in high-level features and transfers these scores to adjacent low-level features to guide the filling of missing regions. Li et al. [?] develop knowledge consistent attention module, which shares the attention scores between recurrences and combines them to guide a patch-swap process. Considering the semantic inconsistencies of contextual attention, Liu et al. [?] propose a coherent semantic attention layer, which additionally considers the correlation between features of missing and repaired regions. Deng et al. [?] propose Laplace attention with spatial positional awareness, which takes into account not only the features themselves but also distance between features.

Some auxiliary information such as edges [?], segmentation maps [?], and gradients [?], which express structural and textural features of images, are beneficial to image inpainting. Specifically, Nazeri et al. [?] propose EdgeConnect, which first repairs the image edges and

subsequently fills in the missing regions based on these edges as a prior. CTSDG [?] introduces a novel two-stream network for image inpainting, which divides the inpainting task into structure-constrained texture synthesis and texture-guided structure reconstruction. Liao et al. [?] propose a Semantic Guidance and Evaluation Network (SGE-Net) which iteratively updates the semantic segmentation maps and utilizes them as guidance in each scale of inpainting. Yang et al. [?] propose a multi-task learning framework, which incorporates auxiliary information including image edges and gradients to assist in image inpainting.

However, the aforementioned methods [?] all adopt static convolution kernels, whose parameters remain fixed once the network is trained, regardless of inputs. This leads to unsatisfactory inpainting results when processing corrupted images with complex scenes. To address this issue, JPGNet [?] and MISF [?] introduce dynamic convolutions, enabling the convolution kernels to dynamically adjust their parameters to accommodate complex and diverse scenes during the testing phase. But they also suffer from shortcomings, as detailed in Section 1. Compared with them, our method fuses multi-scale features to predict attention weights for instantiating ODConv kernels [?] and performs region-wise convolutions to integrate long-range correlations.

3. Proposed Method

In this section, we introduce our image inpainting method called BNRDC (Bi-branch Network with Region-wise Dynamic Convolution). As shown in Figure 1, the pipeline of BNRDC consists of the upper and lower branches. Dynamic Convolution Kernels Prediction (DCKP) and Region-wise Dynamic Convolution Attention (RDCA) are located in the upper and lower branches, respectively, and Multi-Scale Feature Fusion (MSFF) is positioned between the two branches. Given corrupted image I_m and corresponding mask M (with value 1 for missing regions, 0 otherwise), MSFF adaptively fuses the multi-scale features from the lower branch and provides them for DCKP. Based on the features provided by MSFF, DCKP predicts two sets of attention weights to instantiate ODConv kernels [?] for RDCA. RDCA separately convolves the features of missing and complete regions with these ODConv kernels. Subsequently, RDCA borrows features from complete regions to fill in missing regions.

3.1. Multi-Scale Feature Fusion (MSFF)

MSFF fuses the multiple scales feature maps from the lower branch encoder. The fused feature map is then delivered to DCKP, providing semantic feature for the prediction of dynamic convolution kernels. The inputs of MSFF are the feature maps $F_l \in \mathbb{R}^{C_l \times H_l \times W_l}$, $l \in \{1, \dots, L\}$, where F_l denotes the feature map output by the l -th layer of the lower branch encoder, and H_l , W_l , C_l represent the height, width, and channel of F_l , respectively. MSFF generates $\bar{F}_l$ through Feature Enhancement (FE) and Pyramid Feature Fusion (PFF).

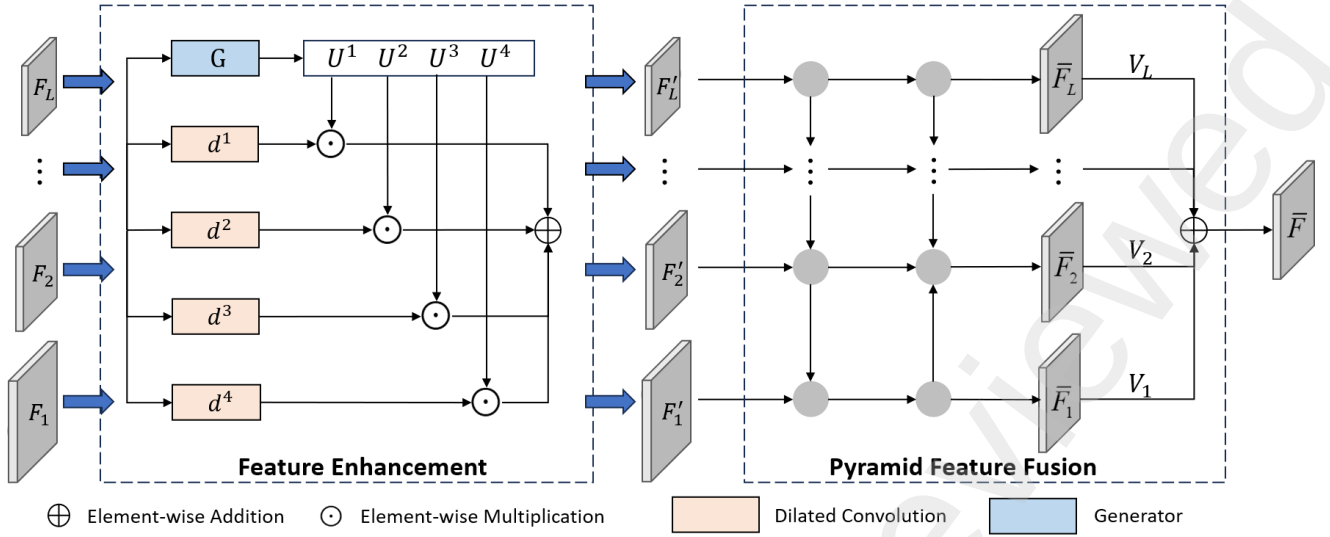


Figure 2: Illustration of the Multi-Scale Feature Fusion (MSFF) module, which fuses the multi-scale features from the lower branch encoder through Feature Enhancement (FE) and Pyramid Feature Fusion (PFF).

Feature Enhancement (FE). FE enhances the feature maps $F_l, l \in \{1, \dots, L\}$ in turn through dilated convolutions and feature weighting, formulated as:

$$\mathbf{F}'_l = \bigoplus_{t=1}^T (\text{Conv}(\mathbf{F}_l, d_l^t) \odot \mathbf{U}_l^t) \quad (1)$$

where $l \in \{1, \dots, L\}$, $\oplus$ and $\odot$ indicate element-wise addition and element-wise multiplication respectively, and T represents the number of dilated convolutions. $\text{Conv}(\mathbf{F}_l, d_l^t)$ stands for dilated convolution with dilation rate d_l^t . We design a weight map generator G , consisting of two convolution layers, to generate a pixel-wise weight map $\mathbf{U}_l^t \in \mathbb{R}^{H_l \times W_l}$. The enhanced feature maps $\mathbf{F}'_l$ are fed into following PFF for semantic and texture interaction.

Pyramid Feature Fusion (PFF). PFF interacts semantic and texture information between feature maps $\mathbf{F}'_l$ through a feature pyramid. The feature pyramid consists of both top-down and bottom-up pathways, along with feature aggregation. Without loss of generality, the feature fusion process on both pathways is expressed as:

$$\mathbf{F}_l^{\text{out}} = \text{Conv} \left(\frac{\alpha \cdot \mathbf{F}'_l + \beta \cdot \text{Resize}(\mathbf{F}_{l+1}^{\text{out}})}{\alpha + \beta + \epsilon} \right) \quad (2)$$

where $\text{Resize}(\cdot)$ denotes upsampling operation on top-down pathway and downsampling operation on bottom-up pathway, and $\text{Conv}(\cdot)$ is a 3×3 convolution operation. α and β are two learnable scalar weights which represent the importance of different inputs, and $\epsilon = 0.0001$ is a small value to avoid numerical instability. As shown in Figure 2, the feature pyramid generates feature maps $\bar{\mathbf{F}}_l$ through two pathways. Prior to feature aggregation, the resolution of the feature maps $\bar{\mathbf{F}}_l$ is scaled to a uniform size. The feature maps $\bar{\mathbf{F}}_l$ are aggregated to produce the refined feature map $\bar{\mathbf{F}}$, expressed as: $\bar{\mathbf{F}} = \bigoplus_{l=1}^L (\bar{\mathbf{F}}_l \odot \mathbf{V}_l)$, where $\mathbf{V}_l$ are spatial weight maps predicted by a weight generator similar to G in FE.

3.2. Dynamic Convolution Kernels Prediction (DCKP)

DCKP aims to predict dynamic convolution kernels for RDCA in the lower branch. Compared with the static convolution kernel, the dynamic one adjusts its parameters in response to inputs, enabling adaptation to complex scenes. DCKP takes the corrupted image $\mathbf{I}_m$ and its corresponding mask $\mathbf{M}$ as inputs. Furthermore, the feature map $\bar{\mathbf{F}}$ generated by MSFF is fed into DCKP to refine the features during the prediction of dynamic convolution kernels. Conditioned on the inputs including $\mathbf{I}_m, \mathbf{M}$ and $\bar{\mathbf{F}}$, DCKP predicts two dynamic convolution kernels, denoted as $\mathbf{K}_m$ and $\mathbf{K}_c$, where the subscripts m and c are abbreviations for “missing” and “complete”, respectively. $\mathbf{K}_m$ and $\mathbf{K}_c$ are dedicated to processing features of missing and complete regions, respectively (see Section 3.3 for details). In this paper, we adopt the Omni-dimensional Dynamic Convolution (ODConv) [?] to predict $\mathbf{K}_m$ and $\mathbf{K}_c$.

The ODConv [?] dynamically adjusts the parameters of the convolution kernel by linearly combining multiple convolution kernels. Given multiple convolution kernels, the corresponding kernel space has four dimensions regarding the spatial kernel size, the convolution kernel number, the input channel number and the output channel number for each convolution kernel. The ODConv employs a SE-typed attention module [?] with four heads to learn attention weights along all four dimensions of the kernel space in a parallel manner. Subsequently, the ODConv combines convolution kernels based on these attention weights, thereby achieving the dynamism of the convolution kernel. Some methods [?] also introduce dynamic convolution to image inpainting, but they need to predict all parameters of the dynamic convolution kernel. In contrast, DCKP only needs to predict several attention weights, instead of all parameters, to instantiate the dynamic convolution kernel.

Typically, there are differences in the feature distributions between the missing and complete regions, thus DCKP separately predicts two dynamic convolution kernels $\mathbf{K}_m$ and $\mathbf{K}_c$. Specifically, the feature map $\tilde{\mathbf{F}}$ output by the DCKP encoder is split into two separate maps through the mask $\mathbf{M}$: $\tilde{\mathbf{F}}_m = \tilde{\mathbf{F}} \odot \mathbf{M}$, $\tilde{\mathbf{F}}_c = \tilde{\mathbf{F}} \odot (\mathbf{1} - \mathbf{M})$, where $\tilde{\mathbf{F}}_m$ and $\tilde{\mathbf{F}}_c$ correspond to the missing and complete regions of $\tilde{\mathbf{F}}$, respectively. The ODConv [?] takes $\tilde{\mathbf{F}}_m$ (and $\tilde{\mathbf{F}}_c$) as input and computes attention weights associated with the four dimensions of the kernel space and these determined weights subsequently instantiate $\mathbf{K}_m$ (and $\mathbf{K}_c$). The following RDCA receives $\mathbf{K}_m$ and $\mathbf{K}_c$ to process features of missing and complete regions, respectively.

3.3. Region-wise Dynamic Convolution Attention (RDCA)

As shown in Figure 1, RDCA, which is the core component of the lower branch, aims to fill in the missing regions at the feature level. Although RDCA is consistent with conventional contextual attention [?] in spirit, the novelties of RDCA lie at the following two points. First, RDCA configures two convolution kernels, namely $\mathbf{K}_m$ and $\mathbf{K}_c$, instead of a single kernel [?], to separately process the missing and complete features of a given feature map. Second, both $\mathbf{K}_m$ and $\mathbf{K}_c$ are dynamic convolution kernels, rather than the static ones used in [?]. These kernels assist RDCA in accommodating variations in feature distributions between different regions.

RDCA partitions the feature map $\mathbf{F}$, produced by the lower branch encoder, into two distinct maps: $\mathbf{F}_m$ and $\mathbf{F}_c$, guided by the mask $\mathbf{M}$. It calculates the cosine similarity s_{ij} between patches extracted from $\mathbf{F}_m$ and $\mathbf{F}_c$, formulated as:

$$s_{ij} = \langle \frac{[\mathbf{F}_c \otimes \mathbf{K}_c]_i}{\|[\mathbf{F}_c \otimes \mathbf{K}_c]_i\|}, \frac{[\mathbf{F}_m \otimes \mathbf{K}_m]_j}{\|[\mathbf{F}_m \otimes \mathbf{K}_m]_j\|} \rangle \quad (3)$$

where $\otimes$ denotes the convolution operation, $\langle \cdot, \cdot \rangle$ represents the cosine similarity, and $[\cdot]_i$ refers to the i -th patch of the feature map, $i \in \{1, 2, \dots, I\}$, $j \in \{1, 2, \dots, J\}$. In Equation 3, the subscripts "m" and "c" signify that RDCA configures two convolution kernels $\mathbf{K}_m$ and $\mathbf{K}_c$ to process $\mathbf{F}_m$ and $\mathbf{F}_c$, respectively. Notably, these kernels $\mathbf{K}_m$ and $\mathbf{K}_c$ are dynamic, which are predicted by DCKP in Section 1. RDCA normalizes s_{ij} to derive the attention score α_{ij} for each patch: $\alpha_{ij} = \exp(s_{ij}) / \sum_{i=1}^I \exp(s_{ij})$. The j -th patch of the missing regions is updated by $\sum_{i=1}^I \alpha_{ij} [\mathbf{F}_c \otimes \mathbf{K}_c]_i$, where $j \in \{1, 2, \dots, J\}$. The low branch decoder receives the updated feature map to generate the inpainting result $\mathbf{I}_{out}$. To further refine details, the dynamic convolution kernel $\mathbf{K}$ output by the upper branch decoder, is applied to convolve with $\mathbf{I}_{out}$.

3.4. Loss Functions

We adopt reconstruction loss, perceptual loss [?], style loss [?], and adversarial loss [?] to train our model.

Reconstruction Loss. The reconstruction loss measures the difference between the inpainting result $\mathbf{I}_{out}$ and the

corresponding ground truth image $\mathbf{I}_{gt}$ at the pixel level, formulated as:

$$\mathcal{L}_{rec} = \mathbb{E} \left[\left\| \mathbf{I}_{out} - \mathbf{I}_{gt} \right\|_1 \right] \quad (4)$$

Perceptual Loss. Since the reconstruction loss struggles to capture high-level semantics, we introduce the perceptual loss $\mathcal{L}_{perc}$ to measure the feature difference between the inpainting result $\mathbf{I}_{out}$ and the ground truth image $\mathbf{I}_{gt}$. The perceptual loss is computed by:

$$\mathcal{L}_{perc} = \mathbb{E} \left[\sum_i \left\| \phi_i(\mathbf{I}_{out}) - \phi_i(\mathbf{I}_{gt}) \right\|_1 \right] \quad (5)$$

where $\phi_i(\cdot)$ is the activation map of the i -th layer of the pre-trained VGG-19[?] on ImageNet [?]. In our work, the subscript "i" stands for *ReLu1_1*, *ReLu2_1*, *ReLu3_1*, *ReLu4_1* and *ReLu5_1* of the VGG-19.

Style Loss. Typically, the transposed convolution layer of the decoder tends to generate checkerboard artifacts [?]. The style loss is introduced to reduce these artifacts, and it calculates the difference between the Gram matrices [?] of the feature maps:

$$\mathcal{L}_{style} = \mathbb{E} \left[\sum_i \left\| \psi_i(\mathbf{I}_{out}) - \psi_i(\mathbf{I}_{gt}) \right\|_1 \right] \quad (6)$$

where $\psi_i(\cdot) = \phi_i(\cdot)^T \phi_i(\cdot)$ denotes the Gram matrix [?] which represents the style of the image.

Adversarial Loss. Image inpainting is a classical inverse ill-posed problem. There are numerous feasible results for identical missing regions. Follow previous works [?], we introduce adversarial training to guide the model in generating the realistic result. The adversarial loss is calculated by:

$$\mathcal{L}_{adv} = \mathbb{E}_{\mathbf{I}_{gt}} [\log D(\mathbf{I}_{gt})] + \mathbb{E}_{\mathbf{I}_{out}} [1 - \log D(\mathbf{I}_{out})] \quad (7)$$

where D is the PatchGAN discriminator [?] with spectral normalization [?] to stabilize training.

Overall Training Loss. The overall training loss function for our model is defined as the weighted sum of the above-mentioned losses.

$$\mathcal{L} = \lambda_r \mathcal{L}_{rec} + \lambda_p \mathcal{L}_{perc} + \lambda_s \mathcal{L}_{style} + \lambda_a \mathcal{L}_{adv} \quad (8)$$

where λ_r , λ_p , λ_s and λ_a are hyperparameters which balance the contributions of different loss terms.

4. Experiment

4.1. Experimental settings

Dataset. We validate the proposed method on three public datasets, i.e., CelebA dataset [?], Places2 dataset [?], and Dunhuang Challenge dataset [?]. The CelebA dataset comprises more than 180 thousand face images with diverse features, encompassing individuals of various ages and genders. The Places2 dataset consists of more than 1.8

Table 1

Quantitative comparisons of our method with baselines on Places2, CelebA and Dunhuang Challenge datasets. $\uparrow$ means higher is better, and $\downarrow$ means lower is better. **Bold** indicates optimal performance

	Datasets	Places2			CelebA			Dunhuang
	Mask Ratio	0%-20%	20%-40%	40%-60%	0%-20%	20%-40%	40%-60%	Default
PSNR $\uparrow$	EdgeConnect[?]]	30.826	24.203	20.138	35.175	27.761	22.762	36.423
	RFR[?]]	29.492	23.746	19.683	32.882	27.884	23.252	37.473
	CTSDG[?]]	32.090	24.909	20.412	36.742	28.766	22.768	38.022
	JPGNet[?]]	31.256	24.694	20.684	35.724	28.230	23.123	37.229
	MISF[?]]	32.254	24.927	20.536	36.845	28.902	23.625	38.442
	Ours	32.411	25.052	20.629	37.087	29.090	23.821	38.721
SSIM $\uparrow$	EdgeConnect[?]]	0.9451	0.8367	0.6918	0.9679	0.8975	0.7865	0.9635
	RFR[?]]	0.9283	0.8060	0.6511	0.9540	0.8915	0.7884	0.9668
	CTSDG[?]]	0.9521	0.8482	0.7021	0.9733	0.9067	0.7898	0.9722
	JPGNet[?]]	0.9474	0.8457	0.7172	0.9716	0.9086	0.8078	0.9699
	MISF[?]]	0.9551	0.8556	0.7117	0.9744	0.9120	0.8073	0.9734
	Ours	0.9555	0.8564	0.7121	0.9754	0.9149	0.8131	0.9748
L1 $\downarrow$	EdgeConnect[?]]	0.0076	0.0234	0.0490	0.0044	0.0146	0.0339	0.0044
	RFR[?]]	0.0089	0.0265	0.0569	0.0053	0.0137	0.0312	0.0030
	CTSDG[?]]	0.0065	0.0217	0.0483	0.0037	0.0128	0.0344	0.0034
	JPGNet[?]]	0.0074	0.0226	0.0467	0.0042	0.0138	0.0327	0.0039
	MISF[?]]	0.0065	0.0212	0.0464	0.0037	0.0126	0.0303	0.0034
	Ours	0.0063	0.0209	0.0460	0.0036	0.0124	0.0296	0.0033
LPIPS $\downarrow$	EdgeConnect[?]]	0.0567	0.1520	0.2717	0.0375	0.1015	0.1892	0.0478
	RFR[?]]	0.0917	0.2083	0.3376	0.0668	0.1139	0.1827	0.0421
	CTSDG[?]]	0.0675	0.1826	0.3160	0.0412	0.1061	0.2247	0.0378
	JPGNet[?]]	0.0790	0.2049	0.3358	0.0423	0.1171	0.2179	0.0466
	MISF[?]]	0.0430	0.1297	0.2499	0.0280	0.0840	0.1659	0.0331
	Ours	0.0421	0.1274	0.2468	0.0267	0.0814	0.1617	0.0322
FID $\downarrow$	EdgeConnect[?]]	1.773	6.957	19.087	0.412	1.912	6.141	23.663
	RFR[?]]	4.824	17.279	47.433	4.623	2.887	4.966	22.396
	CTSDG[?]]	2.611	14.907	50.397	0.445	3.256	13.008	17.619
	JPGNet[?]]	4.391	22.493	57.323	0.801	5.002	13.387	24.669
	MISF[?]]	1.126	4.545	13.969	0.244	1.015	3.199	15.412
	Ours	1.099	4.256	13.015	0.235	0.983	3.096	13.850

million images under 365 scenes, which is suitable for validating the potential of the model for real world applications. The Dunhuang Challenge dataset originates from Dunhuang Grottoes and consists of 500 images for training, which is a practical application of image inpainting. We test all methods and compute the metrics on the standard test set of the three datasets.

Mask Setup. For the CelebA and Places2 datasets, we employ irregular masks provided by [?] to produce corrupted images, which serve as the targets for inpainting. The corrupted images are classified into three categories i.e., 0%-20%, 20%-40%, 40%-60%, according to the mask ratios (area proportion of missing regions in the input image). The Dunhuang Challenge dataset provides official masks to produce input images [?].

Baselines. We choose the following five methods as baselines for comparison with ours, including EdgeConnect [?], RFR [?], CTSDG [?], JPGNet [?] and MISF [?].

EdgeConnect and RFR are representatives based on auxiliary information and progressive inpainting, respectively. The latter three methods share similarities in architecture and innovation with our method, respectively. CTSDG adopts a two-stream architecture; JPGNet and MISF both integrate dynamic convolutions for image inpainting. For a fair comparison, we directly leverage the pre-trained models provided by the authors when available, otherwise train the models by ourselves using the codes and settings provided by the authors.

Implementation Details. The model is implemented in PyTorch 1.12.0 on an Ubuntu 20.04 system, and training is launched on a single NVIDIA 3090 GPU (24 GB) with a batch size of 14. The model is optimized by Adam optimizer [60] with learning rate 0.0001 and decay rate $\beta_1 = 0$, $\beta_2 = 0.9$. Follow MISF [?], we train the model for about 350,000 iterations on three datasets. For all the datasets, if not stated otherwise, the hyperparameters are set as $\lambda_r = 1$, $\lambda_p =$



Figure 3: Qualitative comparison on CelebA, Places2 and Dunhuang Challenge datasets. From left to right: input corrupted images, ground-truth images, the results of EdgeConnect [?], RFR [?], CTSDG [?], JPGNet [?], MISF [?] and ours. Differences are highlighted in the red dashed boxes and the solid boxes are the corresponding zoomed in results.

0.1 , $\lambda_s = 250$ and $\lambda_a = 0.1$. All images and masks are resized to 256×256 for training and testing.

4.2. Quantitative Comparisons.

We employ common evaluation metrics such as peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) [?], L1, Learned Perceptual Image Patch Similarity (LPIPS) [?] and Fréchet Inception Distance (FID) [?] to measure the quality of inpainting results. The first three metrics are calculated at the pixel level, while the latter two at the feature level. Table 1 lists the evaluation results of all methods on the three datasets. Upon analyzing these results, we derive the following four observations.

1) Our method shows superior results across all datasets. Specifically, under the mask ratio of 0%-20%, our method achieves PSNR scores of 37.084 dB and 32.411 dB on CelebA and Places2 datasets, respectively. Even with the 40%-60% mask ratio, the PSNR scores remain steady at 23.821 dB and 20.629 dB, which indicates that our method can deal with large missing regions. Impressively, on the

Dunhuang Challenge dataset, our method shows superior performance, with scores of 38.721 dB, 0.9748, 0.0033, 0.0322, and 13.850 for the five metrics, respectively. This indicates its potential for real world applications.

2) Compared to all baselines, our method shows an average improvement of 3.1% for PSNR, 1.5% for SSIM, 9.5% for L1, 23.8% for LPIPS, and 48.9% for FID on the three datasets. The obvious increase for FID suggests that our inpainting results are closer to the ground truth images in distribution. The superior performance is attributed to our configuration of two dynamic convolution kernels. Distinct dynamic convolution kernels can better adapt to the differences in feature distributions between different regions.

3) In particular, we further compare with CTSDG [?], which adopts the same two-stream architecture as ours. Under the mask ratio of 0-20%, our method achieves an average improvement of 1%, 0.3%, 2.9%, 36.4%, and 52.5% for the five metrics, respectively. In contrast, the performance improvement is even greater under large masks, being

2.7 times that under small masks. This is because our two branches separately predict and utilize dynamic convolution kernels, which can adapt to different scenes during the testing process.

4) Among all baselines, MISF [?] is the most similar to our method, so we further compare with it. Table 1 shows that MISF performs better than other baselines, suggesting the potential of dynamic convolutions in image inpainting. Compared with MISF, our method attains an average improvement of 0.6%, 0.2%, 2.1%, 2.6%, and 5.1% for the five metrics, respectively. This is because MISF only employs single-scale features to predict the dynamic convolution kernel. In contrast, our MSFF integrates multi-scale features to provide richer information, enabling the predicted dynamic convolution kernels to better process features of missing and complete regions.

4.3. Qualitative Comparisons

Figure 3 shows the inpainting results of different methods on three datasets. For the CelebA and Places2 datasets, the mask ratios of input images are 40%-60%. We select two input images with the highest mask ratios from the Dunhuang Challenge dataset, which are 30% and 29.9%, respectively.

As we can see from Figure 3, the content inpainted by JPGNet [?] appears blurry, for example the eyes of the faces. RFR [?] occasionally produces incorrect content, such as the glasses in the second row of images. Although EdgeConnect [?], CTSDG [?] and MISF [?] can generate the main structure of the missing regions, they suffer from missing content and artifacts, etc. Specifically, as observed from the first two rows, the eyes generated by MISF exist artifacts. As shown in the third and fourth rows, their inpainting content lacks railings and the restored walls are invaded by white window frames.

Compared with the inpainting results of all baselines, ours are complete in content and structure, much closer to the ground-truth images. For example, the eyes generated by our method exhibit complete structures without artifacts. Multi-scale features provide richer information, which makes the predicted dynamic convolution kernel more accurate. Additionally, the structure of the railing we restored is complete. This benefits from contextual attention borrowing features from complete regions to fill in missing regions.

4.4. Ablation Study

In this section, we conduct ablation studies to investigate the effectiveness of different modules, namely MSFF and RDCA. There are three ablation settings: "w/o MSFF", "w/o RDCA" and "w/o MSFF & RDCA". In addition, "Complete" represents the complete model, which serves as the upper bound for comparison with three ablation settings. "w/o MSFF" setting indicates that only single-scale features, rather than multi-scale features, are employed for predicting dynamic convolution kernels. As shown in Table 2, the overall performance decreases after removing MSFF module. In particular, the FID shows reductions of 6%, 7.26%, and 15.3% under the three mask ratios, respectively. Compared

Table 2

Ablation studies of different modules on CelebA dataset. ↑ means higher is better, and ↓ means lower is better.

Settings	Metrics	0%-20%	20%-40%	40%-60%
Complete	PSNR↑	37.087	29.090	23.821
	SSIM↑	0.9754	0.9149	0.8131
	L1↓	0.0036	0.0124	0.0296
	LPIPS↓	0.0280	0.0840	0.1659
	FID↓	0.235	0.983	3.096
w/o MSFF	PSNR↑	36.838	29.001	23.658
	SSIM↑	0.9746	0.9129	0.8097
	L1↓	0.0036	0.0126	0.0300
	LPIPS↓	0.0279	0.0835	0.1649
	FID↓	0.250	1.060	3.657
w/o RDCA	PSNR↑	37.004	29.021	23.725
	SSIM↑	0.9750	0.9138	0.8115
	L1↓	0.0036	0.0125	0.0300
	LPIPS↓	0.0273	0.0825	0.1632
	FID↓	0.238	1.002	3.385
w/o MSFF & RDCA	PSNR↑	36.845	28.902	23.625
	SSIM↑	0.9744	0.9120	0.8073
	L1↓	0.0037	0.0126	0.0303
	LPIPS↓	0.0267	0.0814	0.1617
	FID↓	0.244	1.015	3.199

with multi-scale features, single-scale features provide limited information, resulting in less precise predicted dynamic convolution kernels. "w/o RDCA" setting adopts a single dynamic convolution kernel instead of the two kernels to simultaneously process features of missing and complete regions. The removal of the RDCA module results in average reductions of 0.74% and 2.28% for the five metrics under the mask ratios of 0%-20% and 40%-60%. This suggests that a single kernel struggles to adapt to the differences in feature distributions between missing and complete regions. "w/o MSFF & RDCA" setting removes both the MSFF and RDCA modules, resulting in our method degrading to MISF[?]. Most of the metrics exhibit the worst performance when compared to the other settings. This further indicates the effectiveness of the MSFF and RDCA modules.

4.5. Visualization Study

In this section, we conduct two visualization studies. First, we analyze and visualize whether RDCA has the ability to repair the features of missing regions. Second, we visualize and analyze the activation maps under different attention weights of the ODConv kernels [?] for missing and complete regions.

Visualization for RDCA. We randomly sample 1000 pairs of images, consisting of a corrupted image and its ground-truth image, from the CelebA dataset (test set) under the three mask ratios, respectively. For ground-truth images, the features extracted before RDCA are called ground-truth features; For corrupted images, the features extracted before

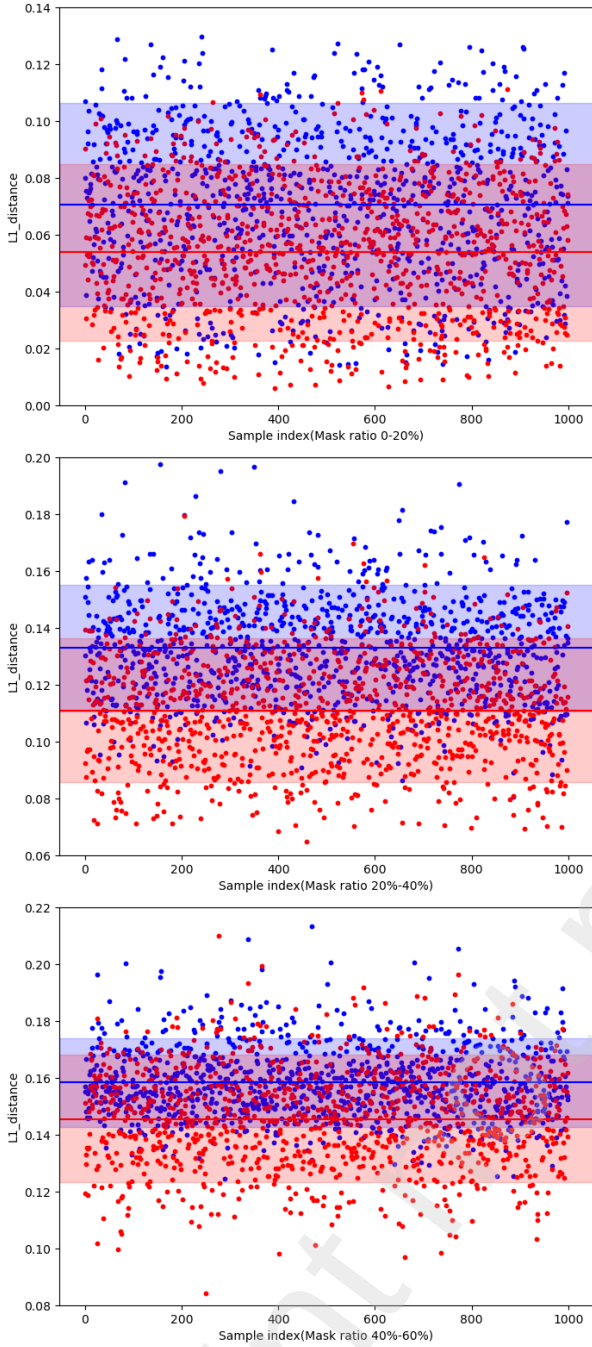


Figure 4: Visualisation of L1 distance between pre-/post- features with ground-truth features. The mask ratios of corrupted images from top to bottom are 0-20%, 20-40%, and 40-60% respectively.

and after RDCA are named as pre-/post-features. We calculate the L1 distance between the pre-features and ground-truth features, obtaining the pre-distance, as shown by the blue dot in Figure 4. Similarly, the post-distance is obtained, represented by the red dot. The smaller the distance, the more similar the features. The lines in Figure 4 represent the averages, and the shaded regions cover the 90% of dots closest to the averages.

As shown in Figure 4, the red lines are always lower than the blue under all three mask ratios. In other words, the post-distances are generally smaller than the pre-distances overall. This indicates that the features processed by RDCA are closer to ground-truth features regardless of mask ratios. As the mask ratio increases, the blue shaded region gradually narrows and moves upward, implying that the pre-distance increases. This is due to that larger missing regions lead to the pre-features gradually deviating from the ground-truth features. In contrast, the red shaded region remains relatively stable. This indicates that even if the area of missing regions increases, RDCA extracts features from different regions employing two dynamic convolution kernels and search for similar features to fill in the missing regions, thereby narrowing the distance between the features of the missing regions and ground-truth images. The aforementioned results suggest that RDCA has the ability to repair the features of missing regions, making them closer to ground-truth features.

Visualization for ODConv kernels. For the four attention weights of the ODConv kernels [?] regarding the input and output channel numbers for each convolution kernel, the spatial kernel size, the convolution kernel number, are denoted as α_i , α_o , α_s and α_k . We compute the activation maps under the joint and individual effects of the four attention weights with Gradient-weighted Class Activation Mapping (Grad-CAM) technique[?], as shown in Figure 5. The first and third rows show the activation maps of the complete regions, while the second and fourth rows display those of the missing regions.

For the ODConv kernels for complete regions, the ODConv kernels with all four attention weights, focus on the areas which are beneficial for repairing missing regions, as shown in the first and third rows of (b). However, under the influence of a single attention weight, the ODConv kernels may focus on areas that are not helpful for repairing missing regions. For instance, the ODConv kernels with α_k and α_k focus on the clothes, as shown in the first row of (e) and (f). For the ODConv kernels for missing regions, the ODConv kernels with all four attention weights, focus on the boundary areas of the missing regions, as shown in the second and fourth rows of (b). Compared to the interior of the missing regions, the boundary areas contain more effective features which are propagated from the complete regions. However, the ODConv kernels with a single attention weight tend to focus on the interior of the missing regions, which contributes less to the repairing the missing regions. From the observations and analysis above, it can be concluded that the ODConv kernels with all four attention weights are more inclined to Concentrate on areas, which are beneficial for repairing the missing regions.

5. Conclusion

In this paper, we propose an image inpainting method called Bi-branch Network with Region-wise Dynamic Convolution (BNRDC), which is composed of the upper and

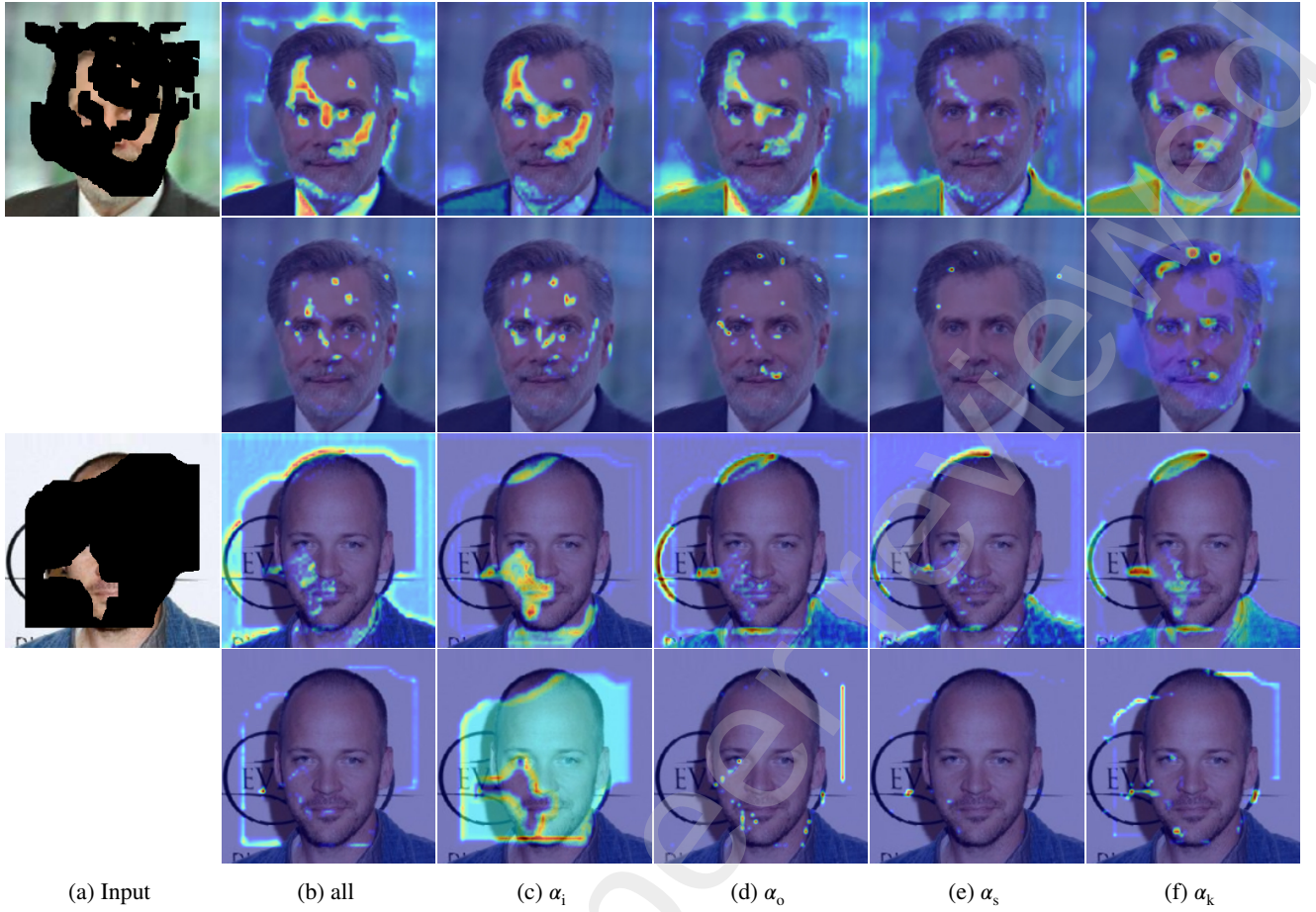


Figure 5: Visualisation of activation maps under different attention weights of the ODConv kernels for missing and complete regions. (a) are the input corrupted images. (b)-(f) are the activation maps under the action of the ODConv kernels with all four attention weights, α_i , α_o , α_s and α_k , respectively.

lower branches. The former is cored with the Dynamic Convolution Kernel Prediction (DCKP) module, and the latter serves as the inpainting backbone. DCKP predicts two sets of attention weights to instantiate the ODConv kernels for the missing and complete regions. To accommodate complex scenes and differences in feature distributions between distinct regions, a Region-wise Dynamic Convolution Attention (RDCA) module is developed to performs region-wise convolutions with two ODConv kernels. Additionally, a Multi-Scale Feature Fusion (MSFF) module is designed to fuse multi-scale features from the lower branch and provide them for DCKP. Extensive experiments on Places2, CelebA and Dunhuang Challenge datasets demonstrate that our proposed method outperforms the state-of-the-art baselines with an average improvement of 0.6%-6.2% for PSNR and 0.2%-3.9% for SSIM, respectively.

Acknowledgements

This work was supported in part by the National Natural Science Foundation of China (62001099), the Fundamental Research Funds for the Central Universities (2232023D-30).